

This exercise sheet provides some training material for the optional tutorial. Some of the exercises and solutions have been taken from Wohlgemuth's book.

Exercise 1

Translate each of the following statements into a first-order formula using $\in$ as the only predicate symbol:

- a) If $A \subseteq B$, then A and $C \setminus B$ are disjoint.
- b) A is not a subset of B .
- c) $\neg(A \cup B \subseteq C \setminus D)$.

Exercise 2

Eliminate def^2 from the proof fragment below.

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| <ol style="list-style-type: none"> 1. $a \in M$ 2. $M \subseteq N$ 3. $a \in N$ $(1,2; \text{def}^2 \subseteq)$ |
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Exercise 3

Let A and B be sets. Define $A \setminus B = \{x \mid x \in A \wedge x \notin B\}$ and $\overline{B} = \{x \in U \mid x \notin B\}$, where U is the universal set. Prove $(A \setminus B) = (A \cap \overline{B})$ in the following two steps.

- a) Show: $(A \setminus B) \subseteq (A \cap \overline{B})$.
- b) Show: $(A \cap \overline{B}) \subseteq (A \setminus B)$.

Exercise 4

Show: $\forall \text{ sets } A \forall \text{ sets } B: ((A \setminus B) \cup B) = (A \cup B)$. Contrary to the examples in the lecture, handle the forall statements on the object level and not on the meta level.

Exercise 5

Show: $\overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \overline{A_i}$ holds for sets $A_1, \dots, A_n$.

Exercise 6

Assume that A , B , and C are sets. Show: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Exercise 7

For sets A , B , and C :

- a) Show: $A \subseteq B$ iff $A \cap B = A$.
- b) Show: $A \subseteq B$ iff $A \cup B = B$.

Exercise 8

Given sets A_k (for $k = 1, 2, 3, 4$), define

$$\begin{aligned} B_1 &= A_1 & B_2 &= A_2 \setminus A_1 \\ B_3 &= A_3 \setminus (A_1 \cup A_2) & B_4 &= A_4 \setminus (A_1 \cup A_2 \cup A_3). \end{aligned}$$

- a) Prove: $\bigcup_{i=1}^4 B_i = \bigcup_{i=1}^4 A_i$
- b) Prove: For all $1 \leq i, j \leq 4, i \neq j: B_i \cap B_j = \emptyset$

Exercise 9

Let $f: A \mapsto B$. Suppose $X \subseteq A$, $Y \subseteq B$, and $y \in B$. Define:

$$\text{P1 } f(X) = \{b \in B \mid b = f(x) \text{ for some } x \in X\}$$

$$\text{P2 } f^{-1}(Y) = \{a \in A \mid f(a) \in Y\}$$

$$\text{P3 } f^{-1}(y) = \{a \in A \mid f(a) = y\}$$

Let $C \subseteq B$, and $D \subseteq B$. Show: $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$. First structure the proof in the same way as we discussed in the lecture. Then present the proof in English.

Exercise 10

Prove or refute the following:

Let $f: A \mapsto B$, $E \subseteq A$, $F \subseteq A$. Then $f(E \cap F) = f(E) \cap f(F)$.

Exercise 11

A function $f: A \mapsto B$ is called *one-to-one* provided that for all $a_1, a_2 \in A$: if $f(a_1) = f(a_2)$ then $a_1 = a_2$.

Let $f: \mathbb{R} \mapsto \mathbb{R}$ be defined by $f(x) = 2x + 4$ for all $x \in \mathbb{R}$. Prove that f is one-to-one.

Exercise 12

A function $f: A \mapsto B$ is called *onto* if for each $b \in B$ there exists some $a \in A$ such that $f(a) = b$.

Show that there exists a one-to-one function $g: \mathbb{N} \mapsto \mathbb{N}$ which is not onto.

Exercise 13

A *sequence* is a function $a: \mathbb{N} \mapsto \mathbb{R}$. $a(n)$ is called the n th term of the sequence and is denoted by a_n . The sequence itself is denoted by $\langle a_1, a_2, \dots \rangle$ or by $\langle a_n \rangle$. A number $L \in \mathbb{R}$ is defined to be the *limit* of $\langle a_n \rangle$, written $\lim_{n \rightarrow \infty} \langle a_n \rangle = L$, provided that, given a real number $\epsilon > 0$, there exists a natural number N such that for all $n > N$: $|a_n - L| < \epsilon$. If L is the limit of $\langle a_n \rangle$, then $\langle a_n \rangle$ *converges* to L . A sequence with no limit is said to *diverge*.

Show that $\langle a_n \rangle$ defined by $a_n = n/(n+1)$ has limit 1.

Exercise 14

Let $\langle a_n \rangle$ be a sequence. If $\lim_{n \rightarrow \infty} \langle a_n \rangle = L$ and $\lim_{n \rightarrow \infty} \langle a_n \rangle = M$, then $L = M$. That is, if a sequence has a limit, then it is unique.

Hint: $|x - z| \leq |x - y| + |y - z|$ for all $x, y, z \in \mathbb{R}$.

Exercise 15

Show: For all integers $n \geq 0$ and $0 \leq i \leq n$, $\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}$ holds.

Hint: $\binom{n}{i} = 0$ if $i < 0$.